Recap: Calculating with Expectations / Variances

$$\left.\begin{aligned} &\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y] \\ &\mathbb{E}[\textstyle\sum_{i=1}^{n} X_i] = \sum_{i=1}^{n} \mathbb{E}[X_i] \\ &\mathbb{E}[aX+b] = a \cdot \mathbb{E}[X] + b \end{aligned}\right\} \text{linearity of expectation (always!)}$$

$$\mathbb{E}_{XY}[g(Y)] = \mathbb{E}_Y[g(Y)] \qquad \left(\int_x \int_y g(y) \cdot p(y,x)\, dy\, dx = \int_y g(y) \cdot \underbrace{\int_x p(y,x)\, dx}_{p(y)}\, dy \right)$$

when $Cov(X,Y) = 0$ (e.g., under independence):

$$\mathbb{E}[X \cdot Y] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$$

Law of total expectation / Tower property:

$$\mathbb{E}_{XY}[\dots] = \mathbb{E}_X \mathbb{E}_{Y|X}[\dots]$$

$$Var(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$Var(aX+b) = a^2 \cdot Var(X)$$

$$Var(X \pm Y) = Var(X) + Var(Y) \pm 2 \cdot Cov(X,Y)$$

For independent variables $X^{(i)}$:

$$Var\left(\textstyle\sum_{i=1}^{n} X^{(i)}\right) = \sum_{i=1}^{n} Var(X^{(i)})$$

when identically distributed:

$$= n \cdot Var(X)$$

Note: $Cov(X,Y) = \mathbb{E}[(X - \mathbb{E}[X]) \cdot (Y - \mathbb{E}[Y])]$

$$= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

independence $\rightarrow Cov(X,Y) = 0$

Law of total variance:

$$Var(Y) = \mathbb{E}_X[Var_{Y|X}(Y)] + Var_X(\mathbb{E}_{Y|X}[Y])$$